The Æther-flow Interpretation of Relativity

Exact-Closure Sequence Overview

Abstract

This note presents the active exact-closure manuscript sequence for Æther-flow Interpretation of Relativity as the flagship public-facing deliverable of the repository. Its purpose is not to replace the individual root manuscripts, but to fix their common role, reading order, and shared claim boundary while packaging them as one coherent theory statement. In the active repository state, Æther-flow Interpretation of Relativity is the exact relativistic theory built on the Æther / Æther-flow ontology. At observable scale, the package is operationally identical to ordinary GR: it keeps the Einstein–Hilbert dynamics, one operative metric, universal matter coupling, ordinary diffeomorphism gauge structure, standard null / proper-time / redshift structure, and no extra preferred-frame, vector, or scalar graviton sector. It is therefore not Einstein-aether theory, not Hořava gravity, and not a new low-energy alternative to GR. Accordingly, the exact-closure sequence already provides a disciplined theory statement at the effective level even though the deeper substrate derivation remains open.

The overview makes four points explicit. First, this note is the canonical front door to the flagship exact-theory package. Second, the exact-closure note and the five companion modules on foundations, dynamics, consistency, relativistic recovery, and flow geometry should be read as one fixed coordinated line after that front door. Third, the predictive content of the active theory is exactly that of GR, while the Æther / Æther-flow vocabulary remains ontological and interpretive at the effective level. Fourth, the separate flagship public article is a release-facing synthesis built on that ordered core, while the derivational bridge manuscripts belong to an ongoing foundational recovery program answerable to this exact-closure benchmark rather than to the completed theory statement itself.

1. Introduction

The active repository has reached a stage at which the exact-closure line should be presented as the flagship scientific deliverable in its own right. The deeper framework still permits ongoing derivational work, but the project is no longer best described as ontology awaiting its first disciplined theory statement. That theory statement now exists in the exact-closure sequence.

At observable scale, that package is ordinary GR in operational content: Einstein–Hilbert dynamics, one operative metric, universal matter coupling, the ordinary ADM / diffeomorphism constraint structure, standard null / proper-time / redshift rules, and no extra preferred-frame, vector, or scalar graviton sector [1, 2]. The congruence-based Æther-flow dictionary belongs to the standard relativistic kinematics literature [3, 4], not to a second observer-level field law. For that reason the active package should be classified up front as not Einstein-aether theory, not Hořava gravity, and not any other low-energy preferred-structure alternative [5, 6, 7, 8, 9, 10, 11]. The remaining derivational ambition belongs instead to the established emergent, induced, reconstruction, self-coupling, and analogue-gravity conversation [12, 13, 14, 15, 16, 17, 18].

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

This distinction is the organizing principle of the overview. The active exact-closure line is already a positive deliverable. The still-open substrate program is a later foundational burden, not the condition for calling the effective theory statement real. The immediate packaging task is therefore consolidation of the flagship exact-theory sequence rather than another derivational bridge installment.

What is new here? What survives as novelty in the current package is interpretive and architectural, not a claim of new low-energy gravitational dynamics. The package enforces a strict adoption-versus-derivation distinction, makes the overview-first benchmark sequence the canonical front door, gives the Æther-flow language a disciplined congruence-based dictionary, and keeps the positive benchmark package separate from the downstream bounded no-go on the frozen derivational line. None of those moves adds a second observer-level field law beyond GR.

What This Package Is Not. For first-time readers, the comparison boundary is deliberately blunt.

- **Not Einstein-aether theory.** The benchmark package does not add a dynamical preferred-frame vector field in the low-energy action; the congruence / Æther-flow layer is a kinematical GR dictionary inside the adopted metric spacetime.
- **Not Hořava or preferred-foliation gravity.** The active package does not claim a preferred slicing, modified low-energy symmetry structure, or an extra low-energy foliation sector.
- **Not a low-energy alternative to GR.** At observable scale the package is ordinary GR with one operative metric and universal matter coupling.
- **Not an already completed emergent-gravity derivation.** The package does not claim that Einsteinian observer dynamics have already been recovered from explicit substrate microphysics or uniquely forced from a substrate action.
- **Not the downstream frozen-line obstruction note.** The current-line no-go is a separate status result on a frozen primitive-reservoir bridge object; it does not define the benchmark package itself.

Positive mirror. The benchmark package is the public result now: an exact-GR theory statement already fixed at observable scale, with the frozen-line no-go kept separate as downstream status and any future derivational restart requiring genuinely new observer-localizing structure.

2. Purpose of the Overview

Definition 1 (Exact-closure sequence overview)**.** *An exact-closure sequence overview is a manuscript that presents the active root papers of Æther-flow Interpretation of Relativity as one coordinated theory statement, identifies the role of each paper within that statement, and fixes the boundary between completed exact closure and unfinished substrate derivation.*

Proposition 1 (Current architectural status)**.** *At the present stage of the repository, the canonical public-facing entry to Æther / Æther-flow framework is the exact-closure manuscript sequence rather than the derivational bridge program.*

This proposition is not merely editorial. It reflects the scientific order now achieved in the repository. The exact operational theory, its consistency status, and its relation to relativistic phenomenology are already stated in disciplined manuscript form. What remains open is not whether the project has any exact theory statement, but whether that exact theory can later be recovered from a more primitive substrate dynamics.

Corollary 1 (Flagship packaging rule)**.** *The fixed flagship release package of the repository is this overview together with the six core exact-closure manuscripts and the public flagship article built on that ordered core, and any further derivational continuation must be presented only as downstream foundational work.*

3. Exact Theory Statement

Definition 2 (Æther-flow Interpretation of Relativity)**.** *Æther-flow Interpretation of Relativity is the exact-closure, GR-consistent theory of Æther / Æther-flow framework in which the Æther / Æther-flow ontology is retained while the effective gravitational dynamics are exactly those of general relativity with universal matter coupling.*

Its effective action is

$$S_{\text{eff}} = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g}\, R + S_{\text{matter}}[g, \psi], \tag{1}$$

with field equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{2}$$

the standard null condition

$$0 = g_{\mu\nu}\, dx^\mu dx^\nu, \tag{3}$$

and the standard proper-time rule

$$d\tau^2 = -\frac{1}{c^2} g_{\mu\nu}\, dx^\mu dx^\nu. \tag{4}$$

Proposition 2 (Operational identity)**.** *For fixed matter sector, initial data, and boundary conditions, the effective observable content of Æther-flow Interpretation of Relativity is identical to that of GR.*

Remark 1. *This is an adoption statement, not a first-principles derivation statement. The exact-closure sequence does not claim that the Einstein sector has already been recovered uniquely from deeper substrate microphysics.*

4. Flagship Package and Reading Order

This note is the canonical front door to the exact-closure sequence. After it, the active core manuscripts should be read in the following fixed order and should be presented in this same order in repository-facing summaries.

1. **Exact-closure note.** The standalone exact-closure note states the shortest complete version of the active scientific position, fixes the benchmark role of Æther-flow Interpretation of Relativity, and marks the nonclaims that preserve discipline.

2. **Foundations.** The foundations manuscript fixes the ontological vocabulary, clarifies the original intent of the framework, and states why exact closure rather than unfinished deformation is the correct active stance.

3. **Dynamics.** The dynamics manuscript states the adopted effective action, field equations, matter-coupling rule, weak-field structure, and benchmark role of the exact-closure theory.

4. **Consistency.** The consistency manuscript makes explicit the gauge structure, degree-of-freedom count, and effective health conditions inherited from GR in the adopted exact-closure theory.

5. **Relativistic recovery.** The relativistic-recovery manuscript states the exact relation to GR and the local relation to SR, including weak-field recovery, null propagation, redshift, clock behavior, and local inertial structure.

6. **Flow geometry.** The flow-geometry manuscript supplies the disciplined congruence-based dictionary by which the Æther-flow ontology is read inside exact relativistic geometry without introducing a second low-energy law.

Corollary 2 (Stable sequence logic)**.** *Read in that order, the exact-closure papers supply one continuous argument: ontology is stated, exact dynamics are fixed, internal consistency is controlled, relativistic recovery is made explicit, and the Æther-flow language is given formal geometric discipline.*

This fixed reading order is part of the claim boundary of the flagship package. It prevents the effective theory statement from being blurred together with optional derivational continuation. The flagship public article sits on top of this ordered core as a release-facing synthesis for first-time readers. It does not replace the overview as the canonical scientific front door, and it does not alter the reading order fixed above.

5. What the Sequence Establishes

The exact-closure sequence now establishes the following at the active theory level.

1. The ontological vocabulary of Æther, Æther-flow, observed three-dimensional space, and S-time is fixed in disciplined form.

2. The effective gravitational law is fixed exactly as Einsteinian with universal matter coupling.

3. The active theory carries no independent low-energy non-GR observable signature.

4. The effective gravitational sector has the ordinary massless spin-2 structure, standard relativistic causal order, and no additional established low-energy scalar or vector graviton sector.

5. The observer-accessible weak-field, redshift, time-dilation, null-propagation, and local SR content is inherited exactly from the same metric structure.

6. The Æther-flow interpretation is formalized through congruence geometry rather than loose metaphor or a naive vector-wind picture.

Proposition 3 (Working theory standard)**.** *At the effective level, Æther-flow Interpretation of Relativity already satisfies the working standard required for the project's main goal: ontology is matched by closed mathematical structure, effective dynamics are explicit, internal consistency is controlled, and observable relativistic structure is recovered without ambiguity.*

6. What the Sequence Does Not Establish

The same discipline requires an explicit statement of what the consolidated exact-closure line does not yet claim.

1. It does not claim a completed first-principles derivation of Einsteinian gravity from explicit substrate variables.

2. It does not claim a unique underlying substrate action from which the exact relativistic sector has already been proved to emerge.

3. It does not claim a distinct verified low-energy observable sector beyond GR.

4. It does not claim that the broader bridge program has already solved uniqueness, mode control, or infrared emergence in full generality.

5. It does not claim that ontology by itself constitutes empirical evidence for new low-energy physics.

Remark 2. *These nonclaims are not weaknesses of the overview. They are part of what makes the exact-closure sequence scientifically stable. The project now succeeds as an exact theory statement because it distinguishes completed effective structure from unfinished microphysical ambition.*

7. Literature Positioning

The benchmark package should not be misread as if Æther / Æther-flow language automatically made it a new low-energy alternative to GR. Its place relative to nearby literatures is narrower and should be stated directly.

1. Relative to standard GR and its congruence / ADM baselines [1, 3, 4, 2], the benchmark package keeps the same low-energy equations and observable content. The difference is interpretive ontology and package discipline, not a second low-energy field law. On the same logic, the consistency manuscript is an audit of the adopted GR sector against the ordinary ADM baseline, not a separate low-energy dynamics discovery.

2. Relative to Einstein-aether theory, static post-Newtonian preferred-frame variants, and Hořava-type preferred-foliation gravity [5, 6, 7, 8, 9, 10, 11], the package is not a preferred-structure modification with a surviving observer-level vector or foliation field and extra propagating modes.

3. Relative to thermodynamic, induced, reconstruction, self-coupling, uniqueness, and analogue-gravity programs [12, 13, 14, 15, 16, 17, 18], the package is closer in spirit only at the level of

the downstream derivation ambition: those programs ask how Einsteinian gravity emerges from deeper structure, just as the open derivational line here asks a related foundational question.

Accordingly, the most accurate current literature label is: exact-GR interpretive package plus open emergent-gravity research program. Any novelty claim must therefore be narrower than "new gravity theory" and narrower than "the general idea that GR might emerge from deeper microphysics." At current scope it can only lie, if anywhere, in the disciplined packaging of the benchmark package, the benchmark-gated derivation architecture, and the decision to preserve the frozen-line no-go explicitly.

8. Relation to the Derivational Bridge Program

The broader substrate manuscript line remains important, but its role must be read correctly. It is answerable to the exact-closure benchmark already fixed above. Its purpose is to ask whether the exact relativistic sector can be recovered from deeper substrate variables while preserving the same operational content and without generating a surviving deformation of that content.

This means the bridge program is no longer the repository's only route to theory status. It is a secondary foundational research program built downstream of an already stated exact theory. That ordering matters. If a future bridge candidate fails, the scientific status of the project contracts back to Æther-flow Interpretation of Relativity rather than back to ontology alone. If the bridge line is deliberately resumed, it should be introduced explicitly as optional continuation beyond the flagship package rather than as the next required milestone.

At the current routing stage, that downstream status has a sharper operational consequence. The bounded primitive-reservoir line has now been carried through its honest test and frozen in the current-line obstruction / no-go note. On that frozen bridge object, the verdict is `Not Derived On Current Line`: the line preserves adoption plus the recorded observer package, but it does not generate the observer-localizing law needed to recover Einsteinian observer dynamics from explicit substrate structure.

The next honest benchmark-facing move is therefore not another deeper same-output relay below the same frozen object. If no concrete new observer-localizing substrate law is already in hand, the correct action is to stop with that scoped no-go and present the exact-closure package as the positive theory statement already secured. If derivational work is resumed, it must begin as a genuinely new line or bridge-design manuscript that states the added law explicitly: how substrate structure localizes to observer spacetime fields, how the operative metric is generated locally, and how the Einsteinian observer equation together with relativistic recovery are supposed to emerge.

The claim boundary also remains fixed while that foundational continuation proceeds. The benchmark package keeps one operative metric, the ordinary matter route stays unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and stronger first-principles promotion language is not yet licensed.

Proposition 4 (Benchmark and reversion role)**.** *Any admissible future extension of Æther / Æther-flow framework must remain answerable to Æther-flow Interpretation of Relativity as exact benchmark, and failure of such an extension contracts the framework back to Æther-flow Interpretation of Relativity rather than invalidating the exact-closure sequence.*

9. Editorial Use of This Note

This overview is the canonical front-door manuscript for the flagship exact-closure line and should be used whenever a short but complete statement of the active project status is needed. It is especially appropriate when the repository must be introduced at sequence level rather than through one root paper in isolation.

The note does not replace the six core manuscripts. Instead, it fixes how they fit together:

- the exact-closure note remains the shortest standalone anchor;
- the five companion papers remain the full modular statement of foundations, dynamics, consistency, relativistic recovery, and flow geometry;
- the flagship public article remains the release-facing synthesis built on this ordered core and does not replace it;
- the substrate notes remain an open derivational program downstream of the exact-closure benchmark and should be presented only after the flagship package is made explicit.

10. Conclusion

The main project goal is now best represented by the exact-closure sequence taken as one coordinated deliverable. Æther-flow Interpretation of Relativity is no longer merely an ontological suggestion awaiting its first disciplined theory statement. It is already the exact relativistic theory of the active framework at the effective level. What remains open is the deeper question of derivation from substrate first principles, not the existence of an exact operational theory.

Accordingly, the correct next architectural stance is consolidation rather than further blurring. The exact-closure sequence should be treated as the canonical flagship theory statement of the repository, and the current bounded derivation line should be read as a scoped no-go on the frozen bridge object rather than as an invitation to keep descending by same-output relay. Any later derivational continuation must therefore appear only as a clearly subordinate foundational research line and must open with an explicitly stated new observer-localizing law or equivalent new structural principle.

References

[1] A. Einstein, "The Foundation of the General Theory of Relativity," *Annalen der Physik* **49** (1916), 769–822.

[2] R. Arnowitt, S. Deser, and C. W. Misner, "The Dynamics of General Relativity," in *Gravitation: An Introduction to Current Research*, edited by L. Witten (Wiley, New York, 1962), 227–265; reprinted in *General Relativity and Gravitation* **40** (2008), 1997–2027.

[3] A. Raychaudhuri, "Relativistic Cosmology. I," *Physical Review* **98** (1955), 1123–1126.

[4] G. F. R. Ellis and H. van Elst, "Cosmological Models (Cargèse Lectures 1998)," in *Theoretical and Observational Cosmology*, edited by M. Lachièze-Rey, NATO Science Series C: Mathematical and Physical Sciences Vol. 541 (Kluwer, Dordrecht, 1999), 1–116, arXiv:gr-qc/9812046.

[5] T. Jacobson and D. Mattingly, "Gravity with a Dynamical Preferred Frame," *Physical Review D* **64** (2001), 024028.

[6] T. Jacobson and D. Mattingly, "Einstein-Aether Waves," *Physical Review D* **70** (2004), 024003.

[7] C. Eling and T. Jacobson, "Static Post-Newtonian Equivalence of General Relativity and Gravity with a Dynamical Preferred Frame," *Physical Review D* **69** (2004), 064005.

[8] T. Jacobson, "Extended Hořava Gravity and Einstein-Aether Theory," *Physical Review D* **81** (2010), 101502(R).

[9] T. Jacobson, "Undoing the Twist: The Hořava Limit of Einstein-Aether Theory," *Physical Review D* **89** (2014), 081501(R).

[10] T. Jacobson and A. J. Speranza, "Variations on an Aethereal Theme," *Physical Review D* **92** (2015), 044030.

[11] P. Hořava, "Quantum Gravity at a Lifshitz Point," *Physical Review D* **79** (2009), 084008.

[12] T. Jacobson, "Thermodynamics of Spacetime: The Einstein Equation of State," *Physical Review Letters* **75** (1995), 1260–1263.

[13] M. Visser, "Sakharov's Induced Gravity: A Modern Perspective," *Modern Physics Letters A* **17** (2002), 977–992, arXiv:gr-qc/0204062.

[14] S. A. Hojman, K. Kuchař, and C. Teitelboim, "Geometrodynamics Regained," *Annals of Physics* **96** (1976), 88–135.

[15] S. Deser, "Self-Interaction and Gauge Invariance," *General Relativity and Gravitation* **1** (1970), 9–18; reprinted with 2004 note at arXiv:gr-qc/0411023.

[16] D. Lovelock, "The Einstein Tensor and Its Generalizations," *Journal of Mathematical Physics* **12** (1971), 498–501.

[17] C. Barcelo, S. Liberati, and M. Visser, "Analogue Gravity," *Living Reviews in Relativity* **14** (2011), 3.

[18] A. D. Sakharov, "Vacuum Quantum Fluctuations in Curved Space and the Theory of Gravitation," *Soviet Physics Doklady* **12** (1968), 1040–1041 [translated from *Doklady Akademii Nauk SSSR* **177** (1967), 70–71].